ELECTRICAL ENGINEERING DEPARTMENT
COLORADO SCHOOL OF MINES

EENG 385 - Electronic Devices and Circuits
Lab 0 - Introduction to LT Spice
Lab Handout

Outcome and Objectives

The outcome of this lab is to familiarize you with the LT Spice circuit simulator so that you can complete all the simulations required in this course. Through this process you will achieve the following learning objectives:

- How to install LT Spice on your computer.
- How to create a schematic.
- How to perform a time domain simulation of a circuit.
- How to perform a frequency domain simulation of a circuit.

Using LTspice: Installing LTspice

LTspice is a **Simulation Program with Integrated Circuit Emphasis** (SPICE) created at the semiconductor company Linear Technologies (LT). LT Spice is free to download and use. To find and download the most recent version of LTspice, type "LTspice Download" into your favorite search engine. Since Linear Technologies was purchased by Analog Devices, you will be directed to the Analog Devices web site. Download the version for your operating system (Windows or MAC – sorry Linux users) and start the install, see Figure 1.

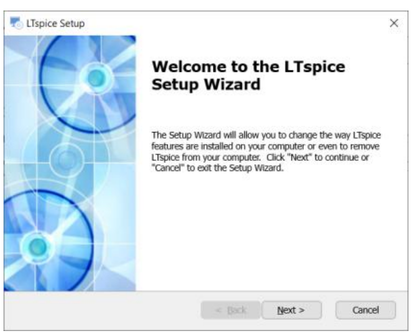
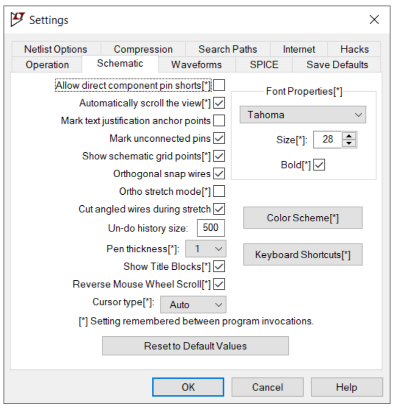


Figure 1: Start screen for the LTspice setup.

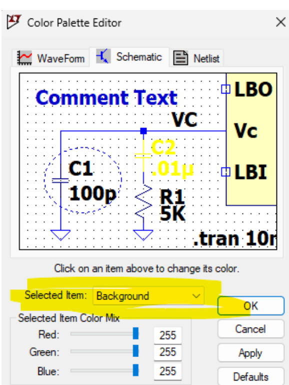
Using LTSpice: Configuring the User Interface

Since you are going to use LTspice a lot in this class, I'd suggest taking a moment of time to configure your user interface to suite your preferences. You can access all the user interface setting by clicking the settings icon at the top of the screen  or selecting Tools -> Settings from the main menu. This will bring up the settings pop-up. With the Schematic tab selected you will see the following.



Background color

My schematics are drawn with a white background. To change the background color, click on the Color Scheme[*] button.



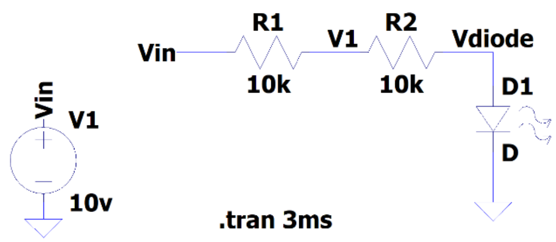
Click Selected Items and choose Background. Set red, green and blue to 255 for white.

Mouse Wheel Zoom

If you are like me, the default mouse wheel/zoom direction relationship will be reversed. In the In the setting pop-up, Schematic tab, check "Reverse Mouse Wheel Scroll".

Using LTspice: Building a Schematic

In your laboratory assignments you will be given two things, a schematic and a table containing simulation parameters like that shown in Figure 2.



Net Names	Vin, V1, Vdiode
Component	LED
Waveform Nets	V1, Vdiode
Configure Analysis	Transient, Stop Time 3ms

Figure 2: The schematic that you will build for this assignment and the table of parameters.

Let's build this circuit and perform the simulation specified. Start by launching LTspice. At the start screen select File -> New Schematic. You should see the schematic canvas in Figure 3; the names of the different tools are shown in red text adjacent to each tool. You will see these names pop-up when you allow your mouse pointer to loiter over a tool. The letter in parentheses next to each tool name is a short-cut key to select that tool without having to point and click on the icon.

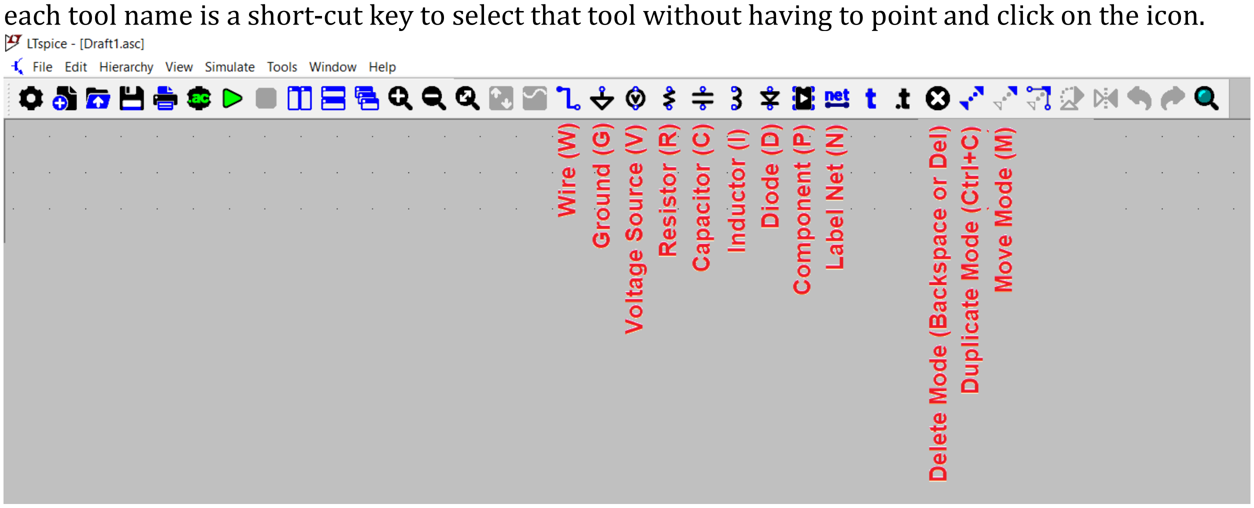


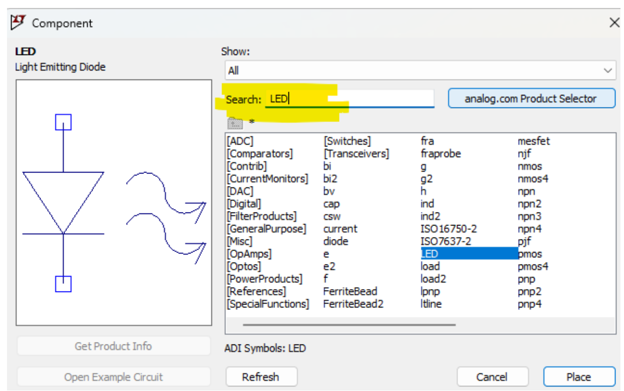
Figure 3: The names of the different components available to build a circuit are shown in red along with their short-cut key. I had not changed the background color to white yet ☹.

The grey area in Figure 3 is the canvas on which you will place your schematic. While building the schematic you will want to move around the schematic. To do this you will:

1) Translate across the schematic by left clicking and dragging your mouse
2) Zoom in/out using the scroll wheel on your mouse.

Step 1: Adding Parts

1) Click on the resistor symbol in the toolbar. Move your cursor into the grey schematic canvas area. The mouse pointer will change into a resistor symbol. Rotate the resistor using Ctrl+R. Place the resistor by left mouse clicking on the schematic canvas. Right mouse click on the resistor will bring up its properties pop-up. Enter “10k” into the top Resistance [Ω] text box.
2) Add a second 10k resistor in the location shown in Figure 4.
3) Click on the ground symbol in the toolbar. Position two of the as shown in Figure 4.
4) Click on the voltage symbol in the toolbar. Position is as shown in Figure 4. After it is placed, right mouse click on the voltage supply to bring up its properties pop-up. Enter “10” into the top DC value[V] text box and then click OK.
5) Click on the component symbol in the toolbar. In the Components pop-up window type “LED” in the Search: textbox. Click Place and the position the LED as shown in Figure 4. In case you have difficulty finding the component check the Component row in the table given with your circuit, Figure 2.



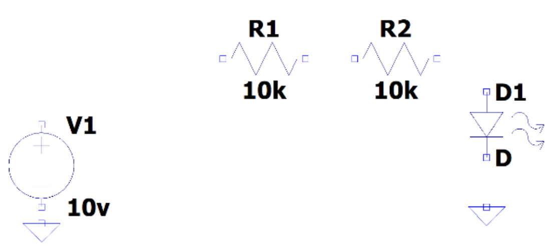


Figure 4: The components placed for the schematic shown in Figure 2.

Try using the Move Mode to move around the components. Do this by typing “m” on your keyboard, then left mouse clicking on the component you want to move, move the component, rotate the component (Ctrl+R), and then left click to place the component in its new location/orientation. You can always use Crtl+Z to undo any action.

Step 2: Adding Wires/Nets

Now you need to use wires to connect everything together. Select the wire tool (or type W) and start connecting the terminals of your placed components so that they look like Figure 5.

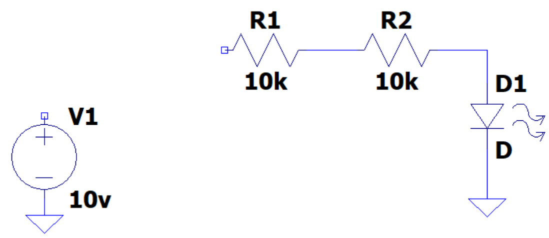
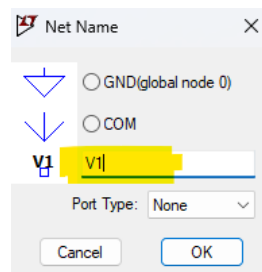


Figure 5: The wires added to the schematic shown in Figure 3.

Step 3: Adding Net Names

Now you need to name the nets so that they are properly named in your waveforms. Select the net tool (or type n) and you should be greeted with the Net Name pop-up. Type "V1" in the textbox and then click on the wire between R1 and R2. You should see "V1" on the wire (really a net) between R1 and R2. Give the wire between R2 and D1 the name "Vdiode"



You can and will use net names to connect signals that are not joined with a drawn wire. This is especially useful when you have a net that connects a lot of different components – drawing all these wires tends to clutter the schematic making the schematic difficult to read.

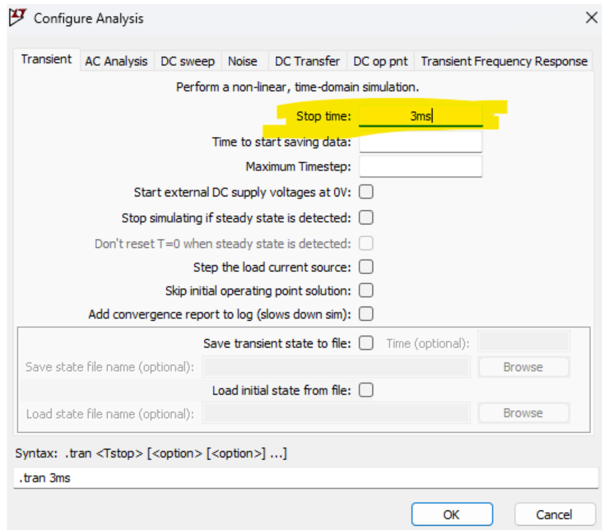
Name the upper terminal the voltage source "Vin" and label the left terminal or R1 "Vin". Since these two terminals are named the same, you have just connected the +10V terminal of the voltage source to the left terminal of resistor R1 – nice work!

Step 4: Export Schematic to Laboratory Report

To show that you have built the schematic for your laboratory report start by saving your work using the save tool or selecting File -> Save from the main menu. Next select Zoom To Fit so that you get the most detail possible in your schematic. Next select Tool -> Copy bitmap to Clipboard from the main menu. Paste the schematic into your lab report using Ctrl+V.

Using LTspice: DC Analysis Simulation

Now that you have created a circuit it's time to simulate it to examine the circuits behavior.



Step 1: Setup Simulation Duration

Start by selecting the Configure Analysis tool from the toolbar at the top of the screen. In the Configure Analysis pop-up enter the stop time given in the table shown in Figure 2 – it was 3ms. When you click OK your cursor will be attached to the text ".tran 3ms". You need to click the mouse and place this text on your schematic so that the simulation will run for 3ms. This command tells the simulator to stop after 3ms.

Step 2: Start Simulation

Now you can run your simulation. Do this by selecting the Run/Pause tool . A waveform window will inject itself into your LTspice window. If you do not see the waveform windows, you have either made an error (you will see a text pop-up), or you need to select the Tile Windows Horizontally tool . If you have an error, please carefully read the text to see what you have missed and fix it.

Step 3: Add Nets to Waveform Window

Before you can see signals in your waveforms window you must select the nets you want displayed. You will be given these signals in the setup table for the circuit (see Figure 2). To get a signal to appear in the waveform window, click on the net in the schematic (your cursor will turn into a red probe) and the net will appear in the waveform window. You can remove a waveform by clicking on the name of the offending waveform at the top of the waveform window and pressing delete/backspace on your keyboard.

Step 4: Export Waveform to Laboratory Report

To show that you have simulated your circuit for your laboratory report start by saving your work using the save tool or selecting File -> Save from the main menu. Next select with waveform window by clicking on its top bar. Then use Tool -> Copy bitmap to Clipboard to save a copy of the waveform that you can paste into your lab report using Ctrl+V.

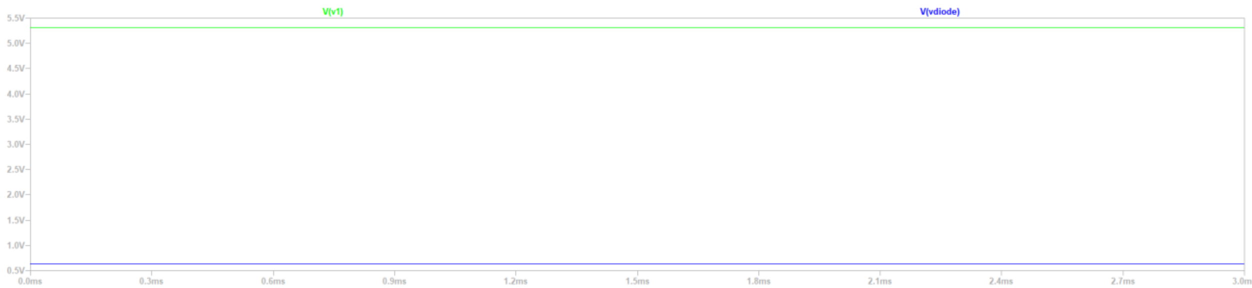
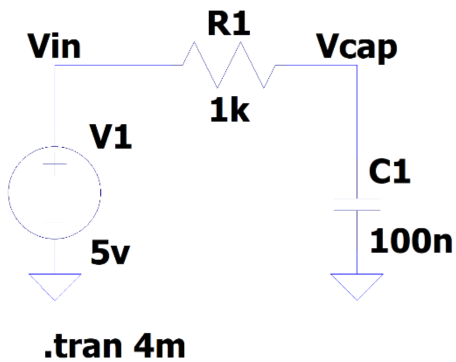


Figure 6: Completed simulation for the circuit shown in Figure 2.

Congratulations, you have finished the DC analysis simulation. Let's move on and look at how

Using LTspice: Voltage Waveform Simulation

Let's look at how to build and analyze circuits with time varying signals. Start by building the schematic in Figure 7. Ignore the Voltage Source row, this will take some explaining.



Net Names	Vin, Vcap
Configure Analysis	Transient, Stop Time 4ms
Voltage Source	5V square wave, 1ms period and 50% duty cycle

Figure 7: Schematic for voltage waveform simulation and analysis.

Let's setup the voltage source V1 according to the specification given in the table above. Just in case some of the terms in the table are new to you, here are some definitions.

- 5V describes the peak to peak amplitude of the square wave. Thus, our waveform will oscillate between 0V and 5V.
- Duty cycle describes the ratio of the time high versus the period, described as a percentage. 50% means that the square wave spends half its time high and half its time low.

Do this by right mouse clicking on the voltage source, in the Voltage Source pop-up, click Advanced. In the Independent Voltage Source pop-up, select the PULSE(V1 V2 … Ncycles) radio button. You will see the pop-up in Figure 8.

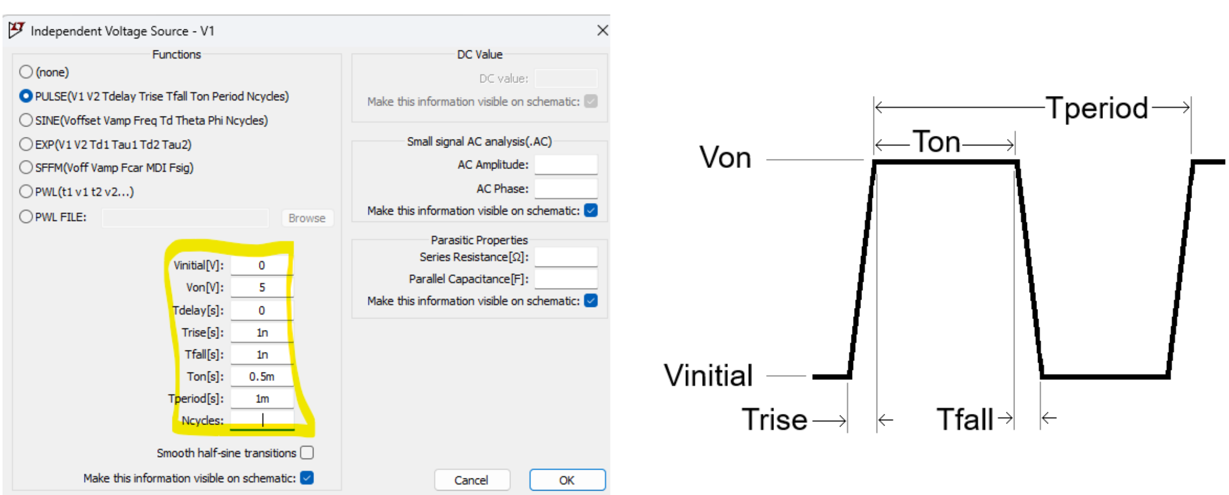


Figure 8: The Independent Voltage Source pop-up used to define the square wave specified in Figure 7.

You will build the square wave by filling in the values circled in yellow in Figure 8. The meaning of the parameters is given in the graph shown in Figure 8, please study and know these. Note, you MUST HAVE A NON-ZERO RISE AND FALL TIME. As a rule of thumb, choose a value that is about 1% of the period. Once you have the square wave parameters entered, click OK and the voltage source will have text "PULSE(0 5 … 1m)" next to it. I like to move this text below the ".tran 4ms" specifier so that all the simulation information is in the same area (making the schematic more readable). Now run your simulation and you should see the timing diagram in Figure 9.

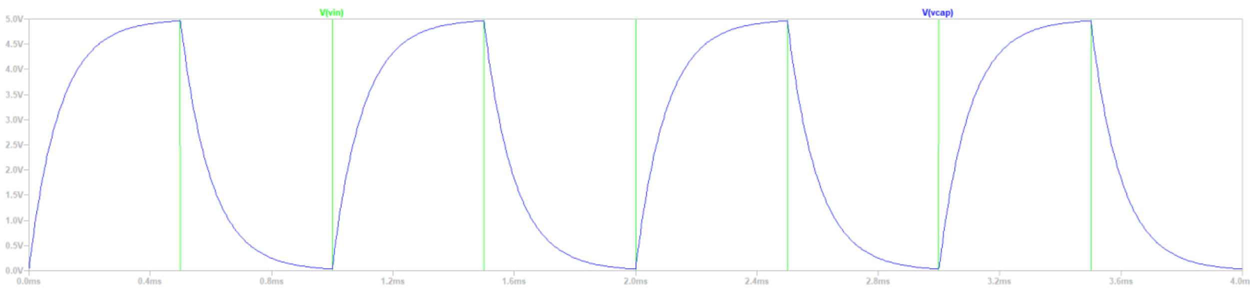


Figure 9: Simulation for 1ms square wave.

Now make the following changes to your simulation:

1) Make Vin a 5V square wave with 100us period and 50% duty cycle,
2) Change the duration of the simulation to 0.8ms,
3) Run with the simulation and probe Vin and Vcap

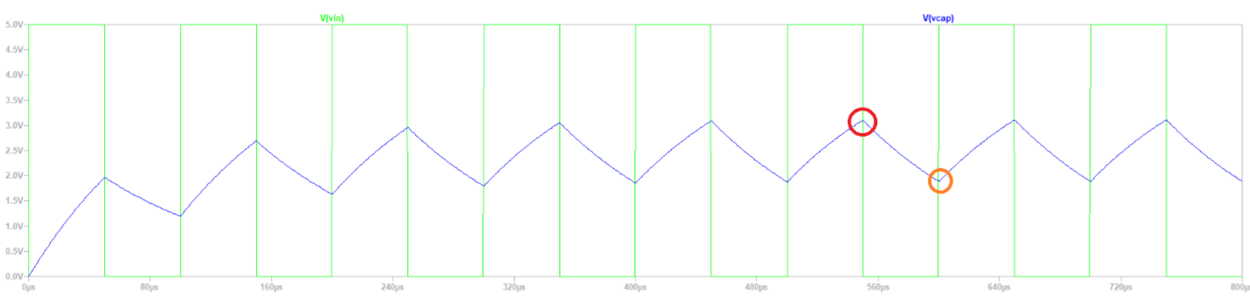


Figure 10: Simulation for 100us square wave.

Note that Vcap never reaches 0V or 5V because the input is changing to quickly. Let's measure the high and low voltages of the Vcap waveform using cursors. Do this as follows:

- Click on the text "V[vcap]" at the top of the waveform window
- Click on Plot Settings -> Place Cursor on Active Trace
- The waveform window will now display a dotted crosshair. Click and drag the crosshair to the peak of Vcap circled in red in Figure 10 while noting the yellow "1" shown while moving the cursor.
- Add a second cursor and position it at the orange circle shown in Figure 10 noting the yellow "2" while moving the cursor.

- All the information about the two cursors is contained in the Draft 1 pop-up shown in Figure 11.

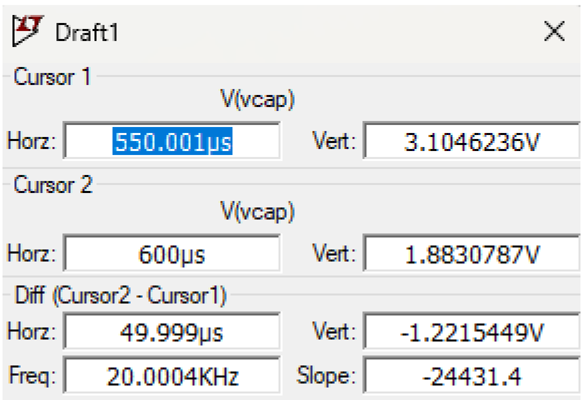


Figure 11: The pop-up associated with cursors provides information about the location of the cursors on the waveform.

Turn In: LTSpice Tutorial

1) Make a record of your response to numbered items below and turn in a single lab report for your team on Canvas using the instructions posted there.
2) Include the names of both team members at the top of your solutions.
3) Use complete English sentences to introduce what each of the items listed below is and how it was derived.

Hint, use Ctrl+click to follow links. This also works for all the Figures and Tables in these labs.

Building a Schematic	Voltage divider schematic
Voltage Waveform Simulation	RC Circuit schematic RC Circuit simulation Compute Vcap peak-to-peak amplitude for 100us wave